

Complex Numbers (Algebra)

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16 August, 2026

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<https://www.youtube.com/@rajatkaliaeducation>

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Blogs

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Part I

Theory

Chapter 1

Complex Numbers

Matrix 1: A line $\bar{a}z + a\bar{z} + c + \bar{c} = 0$ is taken in the argand plane. The equation of line after applying the transformation given, is to be found. Column I contains the transformation while column II contains the resulting equations.

Column I

(P) The line is translated a distance $Re(a)$ along the positive x-axis, $Im(a)$ along positive y-axis, and then reflected in the real axis

(Q) The line is translated a distance $Re(a)$ along the positive x-axis, $Im(a)$ along positive y-axis, and then reflected in imaginary axis

(P) The line is translated a distance $Im(a)$ along the positive x-axis, $Re(a)$ along negative y-axis, and then reflected in the real axis

(P) The line is translated a distance $Im(a)$ along the positive x-axis, $Re(a)$ along positive y-axis, and then reflected in imaginary axis

Column II

(A) $\bar{a}z + az = c + \bar{c} + 2Re(ia^2)$

(B) $\bar{a}z + az + c + \bar{c} = 0$

(C) $\bar{a}z + az + c + \bar{c} - 2|a|^2 = 0$

(D) $\bar{a}z + az + 2|a|^2 = c + \bar{c}$

(E) $\bar{a}z + az + 2Im(a^2) = c + \bar{c}$

Part II

Problem Set

Chapter 2

JEE Mains Problems

Q1: Let x and y be real numbers such that $50 \left(\frac{2x}{1+3i} - \frac{y}{1-2i} \right) = 31 + 17i$, $i = \sqrt{-1}$. Then the value of $10(x - 3y)$ is :

[JEE Mains 2nd Apr 2026 Shift 1 Q2]

(A) 20

(B) 31

(C) 35

(D) 75

{ Solution: <https://www.youtube.com/watch?v=7bvmAsRoeiU> }

Chapter 3

JEE Advanced Problems

Q1: Let $\mathbb{R}$ denote the set of all real numbers. Let $z_1=1+2i$ and $z_2=3i$ be two complex numbers, where $i=\sqrt{-1}$. Let

$$S=\{(x,y)\in\mathbb{R}\times\mathbb{R}|x+iy-z_1|=2|x+iy-z_2|\}.$$

Then which of the following statements is (are) TRUE?

[JEE Advanced 2025 Paper 1 Q7]

(A) S is a circle with centre $(-\frac{1}{3}, \frac{10}{3})$

(B) S is a circle with centre $(\frac{1}{3}, \frac{8}{3})$

(C) S is a circle with radius $\frac{\sqrt{2}}{3}$

(D) S is a circle with radius $\frac{2\sqrt{2}}{3}$

{ Solution : <https://www.youtube.com/watch?v=cYJl2HBn030> }

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